Prediction Field Equation (PFE)

The **Prediction Field Equation (PFE)** formalizes the relationship between prediction capacity, constraint, and informational curvature. It describes how consciousness and information systems participate in spacetime formation through continuous prediction–measurement loops.

General Form:
$P(x, t) = \nabla \cdot [C(x, t) \otimes \Phi(x, t)] - \partial I / \partial t$

where:
- $P(x, t)$ = local prediction field intensity
- $C(x, t)$ = constraint tensor
- $\Phi(x, t)$ = information potential (alignment function)
- I = informational entropy

Constraint corresponds to the Law's first principle ("Constraint is information"), while Φ represents alignment through coordination. The divergence term ($\nabla \cdot$) encodes persistence through iterative correction, establishing spacetime curvature as emergent from predictive coordination.

Interpretation:
When $\partial I / \partial t \rightarrow 0$, prediction equilibrium is achieved — consciousness stabilizes local spacetime geometry.
When $\nabla \cdot C \neq 0$, informational asymmetry introduces potential energy, manifesting as time and motion.
Thus, PFE links quantum measurement, consciousness, and gravitational behavior through predictive necessity.

Summary:
The Prediction Field Equation expresses that prediction creates persistence, constraint defines curvature, and consciousness arises from their coordination. This forms the operational backbone of the *First Signal Law's* predictive cosmology.